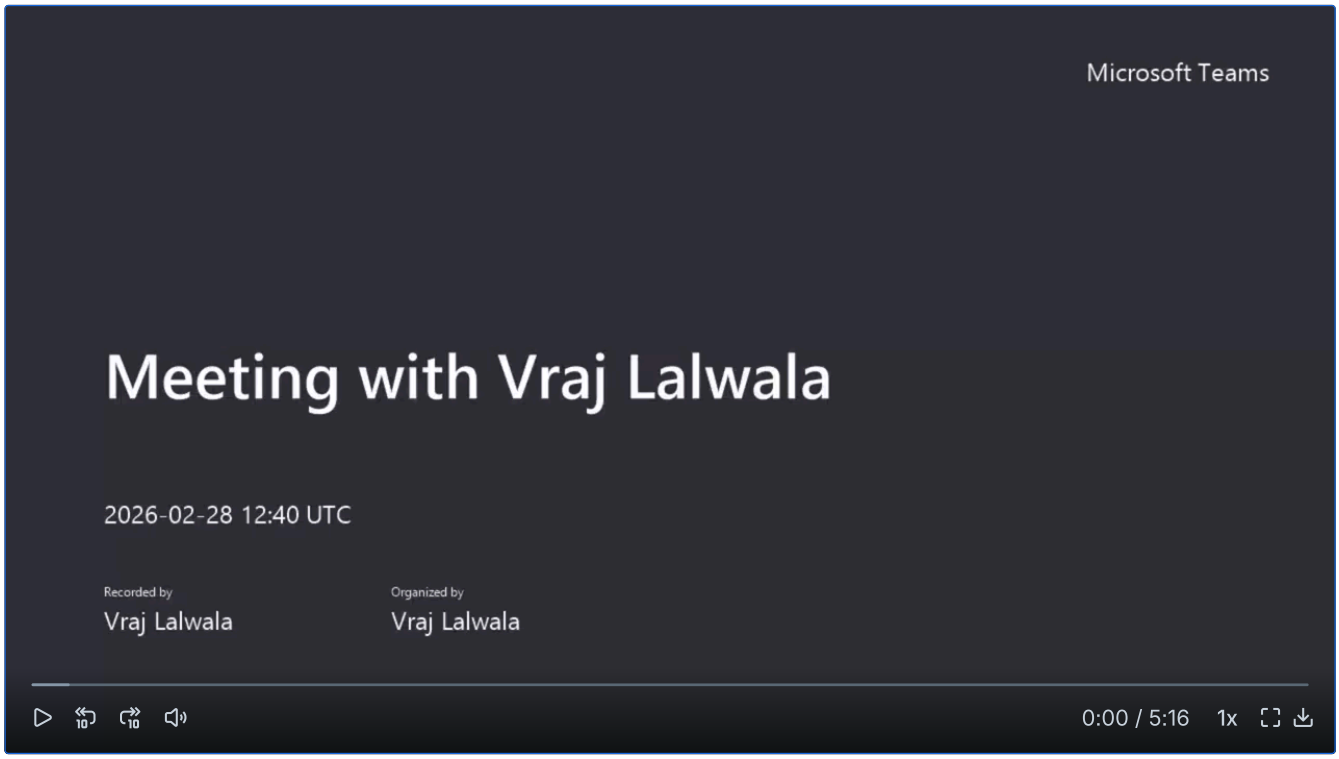
Comparison Report Task-8



Database Monitoring Dashboards – Infrastructure vs Analytics vs Performance Insights

1. Introduction

In modern database environments, monitoring is divided into multiple logical layers.
Each dashboard type serves a different purpose and targets different stakeholders.

This document explains the difference between:

- Infrastructure Dashboard
- Analytics (Application) Dashboard
- Performance Insights Dashboard

Using Amazon RDS PostgreSQL as the reference architecture.

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2. Infrastructure Dashboard

2.1 What It Monitors

Infrastructure dashboards focus on **system-level health and resource utilization** of the database server.

Typical Metrics (from CloudWatch):

- CPU Utilization
- Freeable Memory
- Free Storage Space
- Read IOPS
- Write IOPS
- Network Throughput
- Database Connections

2.2 Data Source

- AWS CloudWatch
- System metrics provided by RDS

2.3 Purpose

To answer questions like:

- Is the DB server overloaded?
- Are we running out of memory?
- Is disk space decreasing rapidly?
- Are IOPS limits being reached?

2.4 Target Audience

- DevOps Engineers
- Cloud Engineers
- Infrastructure Teams
- SRE Teams

2.5 When It Is Used

- Capacity planning
- Scaling decisions
- Resource bottleneck identification

- Cost optimization

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3. Analytics (Application) Dashboard

3.1 What It Monitors

Analytics dashboards focus on **business data stored inside the database**.

Examples:

- Total number of cities
- Total countries
- Population by continent
- Top 10 countries by number of cities
- Application-specific aggregations

These queries run directly on PostgreSQL.

3.2 Data Source

- PostgreSQL datasource in Grafana
- Direct SQL queries

3.3 Purpose

To answer questions like:

- What insights can we extract from the data?
- How is business data distributed?
- What trends exist in the dataset?

3.4 Target Audience

- Product Teams
- Business Analysts
- Data Teams
- Application Owners

3.5 When It Is Used

- Business intelligence
- Reporting
- Data exploration

- Operational analytics

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4. Performance Insights Dashboard

4.1 What It Monitors

Performance Insights dashboards focus on **database engine internals and query behavior**.

Typical Metrics:

- DB Load (Active Sessions)
- Top SQL Queries
- Wait Events
- Query execution time
- Blocking sessions
- CPU time per query

4.2 Data Source

- Amazon RDS Performance Insights
- Query-level monitoring from the DB engine

4.3 Purpose

To answer questions like:

- Which query is causing high CPU?
- Why is DB load increasing?
- Are sessions blocked?
- What wait events are occurring?

4.4 Target Audience

- Database Administrators (DBA)
- Backend Engineers
- Performance Engineers

4.5 When It Is Used

- Query optimization
- Debugging slow performance
- Investigating production incidents

- Load testing validation

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5. Key Differences Summary

Category	Infrastructure Dashboard	Analytics Dashboard	Performance Insights Dashboard
Focus	Server health	Business data	Query-level performance
Data Source	CloudWatch	PostgreSQL	RDS Performance Insights
Level	System-level	Application-level	Engine-level
Audience	DevOps / SRE	Product / BI	DBA / Backend
Problem Solved	Resource bottleneck	Data insights	Query bottlenecks

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6. Practical Example Scenario

If CPU spikes:

1. Infrastructure dashboard shows CPU = 80%
2. Performance Insights shows a specific SQL query consuming 70% CPU
3. Analytics dashboard remains unaffected because business metrics are still correct

This layered monitoring approach allows full-stack visibility.

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7. Why All Three Are Important

Relying on only one dashboard is incomplete:

- Infrastructure alone cannot identify which query is slow.
- Performance Insights alone cannot show business metrics.
- Analytics alone cannot show resource saturation.

Together, they provide:

- Observability
- Root cause analysis

- Business visibility
- Operational reliability

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8. Conclusion

A mature database monitoring setup must include:

1. Infrastructure monitoring (system health)
2. Analytics monitoring (business data visibility)
3. Performance monitoring (query-level insights)

This separation ensures:

- Faster troubleshooting
- Better scalability planning
- Clear stakeholder alignment
- Production-grade observability architecture